# NBTS TEST-6 (Rolex)

1. Choose the incorrect statement(s) from the following

a. The tangent to the magnetic field lines at a given point represents the direction of the net magnetic field $B$ at that point
b. The magnetic field lines of a magnet always form open loops
c. The larger the number of field lines crossing per unit area, the stronger is the magnitude of magnetic field $B$
d. Both (1) and (2)

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2. The transmission of heat through convection can take place in

a. Only liquid
b. Only gases
c. Both solid and liquid
d. Both liquid and gases

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3. The magnitude of the magnetic field at equitorial point due to bar magnet of magnetic moment 0.4 $Am^2$ of length 1 cm at a distance of 50 cm from its midpoint.

a. $3.2 \times 10^{-7}$ T
b. $10^{-7}$ T
c. $1.6 \times 10^{-7}$ T
d. $6.4 \times 10^{-7}$ T

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4. When 1 kg of ice at 0°C is mixed with 2 kg of water at 50°C in a container, the resulting temperature is 7°C. The latent heat of fusion of ice is ($S_{water}$ = 4200 J $kg^{-1}$ $K^{-1}$)

a. $9 \times 10^5$ J/kg
b. $3.3 \times 10^3$ J/kg
c. $3.3 \times 10^6$ J/kg
d. $3.3 \times 10^5$ J/kg

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5. A cup filled with hot chai cools from 90°C to 80°C in 5 minutes when the room temperature is at 20°C. The time it takes to further cool from 80°C to 60°C is

a. 13 minutes
b. 10 minutes
c. 5 minutes
d. 26 minutes

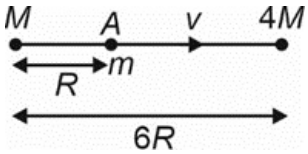
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6. The coefficient of volume expansion of INVAR is $2 \times 10^{-6}$ $K^{-1}$. The fractional change in its density for a 40°C rise in temperature will be

a. $2 \times 10^{-7}$
b. $8 \times 10^{-7}$
c. $2 \times 10^{-5}$
d. $8 \times 10^{-5}$

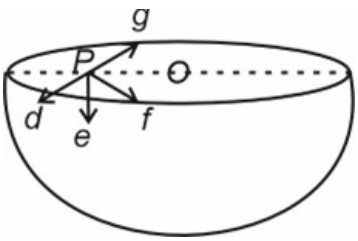
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7. Two fixed particles of masses $M$ and $4M$ have a separation of $6R$, as shown in figure. A particle of mass $m$ is projected from point $A$ towards the mass $4M$. The value of minimum speed $v$ of the projected particle so that it strike the mass $4M$ is




a. $\sqrt{\frac{3GM}{R}}$
b. $\sqrt{\frac{GM}{R}}$
c. $\sqrt{\frac{GM}{5R}}$
d. $\sqrt{3GM}$

8. The direction of gravitational intensity at an arbitrary point $P$ of a hemispherical shell of uniform mass density is indicated by the arrow (O is the centre of hemisphere




a. $d$
b. $e$
c. $f$
d. $g$

a. $\frac{}{5\ R}$

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9. A body weighs 84 N on the surface of the earth. The gravitational force on it due to the earth at a height equal to radius of the earth is

a. 42 N
b. 84 N
c. 21 N
d. 48 N

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10. A structural steel rod has a radius of 1 mm and a length of 1 m. If 100 kN force stretches it along its length, then elongation in rod is (Young's modulus of steel is $2 \times 10^{11}$ $Nm^{-2}$)

a. 0.16 m
b. 1.6 m
c. 2.6 m
d. 16 m

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11. A fully loaded AIR INDIA aircraft has a mass of $4 \times 10^5$ kg. Its total wing area is 500 $m^2$. If it is in level flight with a speed of 260 m/s, then pressure difference between the lower and upper surfaces of the wings is (Take $g = 10$ $m/s^2$)

a. 400 $Nm^{-2}$
b. $4 \times 10^3$ $Nm^{-2}$
c. $8 \times 10^3$ $Nm^{-2}$
d. $10^3$ $Nm^{-2}$

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12. For a paramagnetic material, the dependence of the magnetic susceptibility $\chi$ on the absolute temperature $T$ is given as

a. $\chi \propto T$
b. $\chi \propto \frac{1}{T^2}$
c. $\chi \propto \frac{1}{T}$
d. $\chi \propto T^{\circ}$

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13. A liquid does not wet the surface, if the angle of contact is

a. 90°
b. Acute
c. Obtuse
d. Zero

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14. The working of an atomizer based upon

a. Bernoulli's principle
b. Archimedes principle
c. Newton's law of motion
d. First law of thermodynamics

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15. If a drop of water is broken into smaller drops the surface energy will

a. Increases
b. Decreases
c. Remains unchanged
d. Can increase as well as decreases

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16. The temperature on Celsius scale is 20°C. What is the corresponding temperature on the Fahrenheit scale?

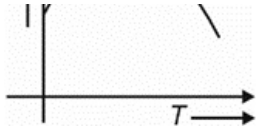
a. 77°C
b. 86°F
c. 68°F
d. 40°F

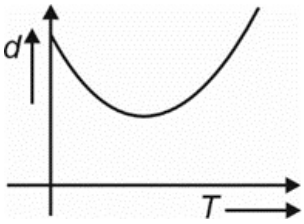
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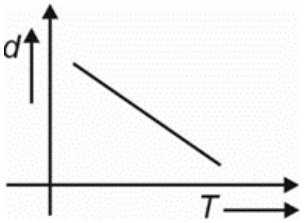
17. The variation of density ($d$) of water with temperature ($T$) is best represented by

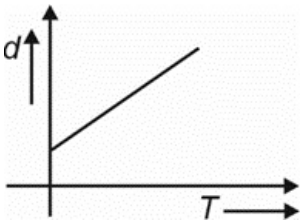



18. If potential at surface of earth is taken as zero, then find potential at centre of earth. (Symbols have usual meanings)??

a. 


b. 


c. 


d. 


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a. $-\frac{3}{2}\frac{GM}{R}$
b. $-\frac{GM}{2R}$
c. $\frac{3}{2}\frac{GM}{R}$
d. $\frac{GM}{2R}$

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19. If energy of a body-planet system is positive, then body will

a. Escape out from planet
b. Move around the planet in elliptical orbit
c. Move around the planet in circular orbit
d. Move to certain maximum height and come back to surface of the planet

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20. Equation of continuity is based on

a. Conservation of volume
b. Conservation of mass
c. Conservation of energy
d. Conservation of linear momentum

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21. Mode of transmission of heat in which heat is carried by the moving particles is

a. Conduction
b. Convection
c. Radiation
d. Both (2) & (3)

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22. A tank is filled to height $H$. The range of water coming out from the hole on side wall at depth H/4 from free surface of water level is

a. $H$
b. $\frac{3H}{4}$
c. $\frac{2H}{\sqrt{3}}$
d. $\frac{\sqrt{3}H}{2}$

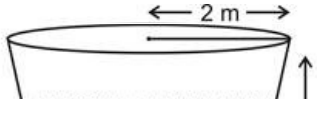
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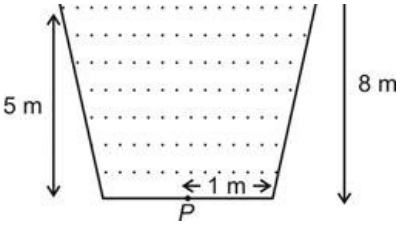
23. In case of a curved surface of a liquid, pressure is greater at

a. Convex side
b. Concave side
c. Equal at both sides
d. Depends on radius of curvature

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24. In an open container water is filled upto a height of 5 m as shown. With the given dimensions of container, the pressure at the bottom of container (at point $P$) is (take atmospheric pressure $1 \times 10^5$ Pa, $g$ = 10 m/s$^2$, density of water 1000 kg/m$^3$)?






a. $1 \times 10^5$ Pa
b. $1.5 \times 10^5$ Pa
c. $1.8 \times 10^5$ Pa
d. $0.5 \times 10^5$ Pa

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25. $g_e$ and $g_p$ respectively denote the acceleration due to gravity on the surface of earth and on another planet whose mass is same as that of the earth and radius is twice that of the earth, then

a. $g_p = \frac{g_e}{4}$
b. $g_p = 4g_e$
c. $g_p = \frac{g_e}{2}$
d. $g_p = g_e$

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26. Needles $N_1$, $N_2$, $N_3$ are made of a ferromagnetic substance, paramagnetic substance and a diamagnetic substance respectively. A magnet when brought close to them will

a. Attract all three of them
b. Attract $N_1$ and $N_2$ strongly but repel $N_3$
c. Attract $N_1$ strongly, $N_2$ weakly and repel $N_3$ weakly
d. Attract $N_1$ strongly, but $N_2$, $N_3$ weakly

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27. Isothermal Bulk modulus of gases is equal to

a. Temperatures of gas
b. Pressure of gas
c. Zero
d. Infinity

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28. The relative permeability ($\mu_r$) of a substance is related to its susceptibility ($\chi$) as

a. $\mu_r = 1 - \chi$
b. $\mu_r = 1 + \chi$
c. $\mu_r = 1 - \chi^2$
d. $\mu_r = 1 + \chi^2$

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29. If mass of earth is changed, then time period of revolution of earth around sun

a. Remains same
b. Increases
c. Decreases
d. Depends, whether mass has been increased or decreased

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30. Find the percentage decrease in the weight of an object when it is taken half way down to the centre of the earth?

a. 40%
b. 10%
c. 30%
d. 50%

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31. Curie temperature is the temperature above which

a. A paramagnetic substance behaves like diamagnetic
b. A ferromagnetic substance behaves like diamagnetic

32. Two soap bubbles of different radii are connected to each other through a tube as shown.

Choose the most appropriate alternative.



c. A ferromagnetic substance behaves like paramagnetic
d. A diamagnetic substance behaves like paramagnetic

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a. Size of the bubbles remains unchanged
b. Larger bubble bursts instantly
c. Air flows from larger bubble into the smaller one
d. Air flows from smaller bubble into the larger bubble

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33. According to Kepler, the law of area can be understood as a consequence of conservation of

a. Linear momentum
b. Angular momentum
c. Mass
d. Energy

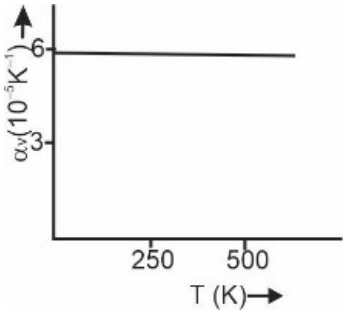
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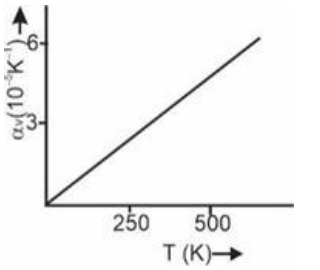
34. A freely suspended magnetic needle is slightly rotated from its equilibrium position in the external magnetic field $B$ and released. Its moment of inertia is $I$ and magnetic dipole moment is $M$, then the least time after which it aligns with the field is

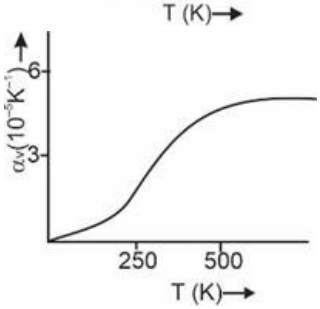
a. $\pi\sqrt{\frac{I}{MB}}$
b. $\pi\sqrt{\frac{I}{8MB}}$
c. $\pi\sqrt{\frac{I}{4MB}}$
d. $\pi\sqrt{\frac{2I}{MB}}$

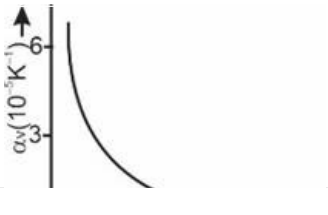
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35. Choose the correct graph of the coefficient of volume expansion of copper.

a.



b.


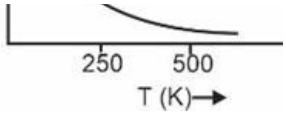

c.



36. Hook's law is valid up to

a. Elastic limit
b. Plastic limit
c. Proportional limit
d. Fracture point

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d. 


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37. The SI unit of stress is same as the SI unit of

a. Strain
b. Modulus of elasticity
c. Pressure
d. Both (2) & (3)

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38. If water is heated from 0°C to 8°C then the volume

a. Increases throughout the temperature
b. Decreases throughout the temperature
c. First decreases then increases
d. Remains same

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39. $\chi_1$ and $\chi_2$ are susceptibility of a paramagnetic substance at temperature $T_1$ and $T_2$ respectively, then

a. $\chi_1 T_1 = \chi_2 T_2$
b. $\chi_1 = \chi_2$
c. $\chi_1 T_2 = \chi_2 T_1$
d. $\chi_1 T_1^2 = \chi_2 T_2^2$

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40. One can easily "weight the earth" by calculating the mass of earth using the formula (symbols have their usual meaning)

a. $\frac{G}{g}R_E^2$
b. $\frac{g}{G}R_E^2$
c. $\frac{g}{G}R_E$
d. $\frac{G}{g}R_E^3$

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41. The kinetic energy needed to project a body of mass $m$ from the earth surface (radius $R$) to infinity is

a. $\frac{mgR}{2}$
b. $2\ mgR$
c. $mgR$
d. $\frac{mgR}{4}$

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42. The radii of two planets are in the ratio 4 : 1 and their mean densities in the ratio 1 : 2. The respective acceleration due to gravity on the planets will be in ratio

a. 3 : 2
b. 2 : 3
c. 1 : 2
d. 2 : 1

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43. The average depth of Indian ocean is about 2000 m. The fractional compression , $\frac{\Delta V}{V}$ of water at the bottom of the ocean is (Bulk modulus of water is $2.2 \times 10^9\ \text{Nm}^{-2}$ and take $g = 10\ \text{ms}^{-2}$)

a. $0.9 \times 10^{-2}$
b. $0.09 \times 10^{-2}$
c. $1.1 \times 10^{-2}$
d. $11 \times 10^{-2}$

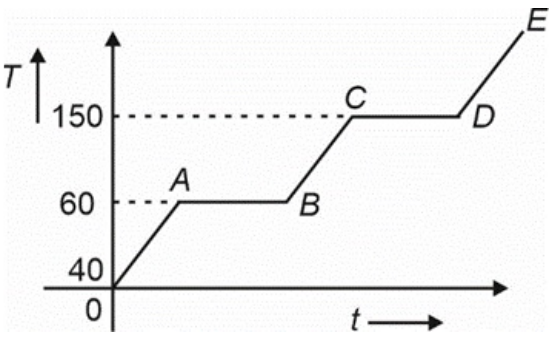
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44. If the molecules of different materials in contact attract each other, then surface energy

a. Decreases
b. Increases
c. Remains unchanged
d. Can increase as well as decreases

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45. A solid substance is at 40°C. To this substance heat energy is supplied at a constant rate. The temperature versus time graph is as shown in the figure. The substance is in liquid state for the portion (of the graph) is




a. *OA*
b. *AB*
c. *BC*
d. *DE*

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46. Gravitational potential in a region is given by $V = -(xyz)$ J/kg. Find gravitational intensity at (2, 2, 2) m.

a. $\left(\hat{i} + \hat{j} + \hat{k}\right)$ N/kg
b. $2\left(\hat{i} + \hat{j} + \hat{k}\right)$ N/kg
c. $3\left(\hat{i} + \hat{j} + \hat{k}\right)$ N/kg
d. $4\left(\hat{i} + \hat{j} + \hat{k}\right)$ N/kg

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47. When two liquids of equal mass but different densities $\rho_1$ and $\rho_2$ are mixed together, then the density of the mixture is??

a. $\frac{\rho_1 + \rho_2}{2}$
b. $\frac{2\,\rho_1\rho_2}{\rho_1 + \rho_2}$
c. $\rho_1 + \rho_2$
d. $2(\rho_1 + \rho_2)$

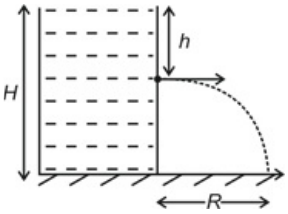
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48. In a capillary of length $2h$, water rises upto height $h$ when it is dipped in water. If the whole system is released to fall freely under gravity, then water will rise upto height

a. $h$
b. $2h$
c. $1.5h$
d. Data is insufficient to predict

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49. A liquid is coming out from the orifice of tank and falls upto a maximum horizontal distance of 10 m. The height $h$ is equal to




a. 5 m
b. 15 m
c. 10 m
d. 7.5 m

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50. A : For a perfectly rigid body value of Bulk modulus is infinite.

R : For a perfectly rigid body change in volume is zero.

a. Both Assertion & Reason are true and the reason is the correct explanation of the assertion
b. Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
c. Assertion is true statement but Reason is false
d. Both Assertion and Reason are false statements

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51. Balance the given redox reaction in acidic medium and match the reactants/products given in column I with their coefficient in column II.

$S_2O_3^{2-} + Br_2 + H_2O \longrightarrow SO_4^{2-} + Br^- + H^+$

**Column I** **Column II**

52. In which of the given compounds, chlorine is in maximum oxidation state?

a. $KClO_3$
b. $ClO_2^-$
c. $HClO_4$

| Column I | Column II |
|---|---|
| a. $S_2O_3^{-2}$ | (i) 10 |
| b. $Br_2$ | (ii) 8 |
| c. $H^+$ | (iii) 4 |
| d. $Br^-$ | (iv) 1 |

a. a(iv), b(iii), c(ii), d(i)
b. a(ii), b(iii), c(iv), d(i)
c. a(iii), b(iv), c(i), d(ii)
d. a(iv), b(iii), c(i), d(ii)

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d. KCl

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53. The number of S = O bond(s), S – O bond(s) and S – S bond(s) respectively in tetrathionate ion are

a. 4, 2 and 3
b. 4, 2 and 1
c. 2, 1 and 3
d. 2, 2 and 1

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54. Choose the incorrect statement.

a. $Pb_3O_4$ is a stoichiometric mixture of 2 mol of PbO and 1 mol of $PbO_2$
b. Reaction of $Pb_3O_4$ with HCl leads to the formation of $PbCl_2$, $Cl_2$ and $PbO_2$
c. Reaction of $Pb_3O_4$ with $HNO_3$ results in the formation of $Pb(NO_3)_2$, $PbO_2$ and $H_2O$
d. Reaction of PbO with HCl is an acid-base reaction

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55. On the basis of the following standard electrode potential values arrange the given metals in the order of their increasing reducing power.

I $K^+/K = -2.93$ V

II $Mg^{2+}/Mg = -2.36$ V

III $Cr^{3+}/Cr = -0.74$ V

IV $Cu^{2+}/Cu = 0.34$ V

a. III < II < I < IV
b. IV < III < II < I
c. IV < II < III < I
d. I < II < III < IV

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56. Consider the following statements and choose the correct option:

**Statement I :** The cathode and the anode in Daniel cell are made up of copper and zinc respectively.

**Statement II :** By convention, the standard electrode potential of standard hydrogen electrode is 1 volt.

a. Both the statements I and II are correct
b. Both the statements I and II are incorrect
c. Statements I is correct but statement II is incorrect
d. Statements II is correct but statement I is incorrect

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57. The element which exhibits only positive oxidation state in its compounds is

a. Cl
b. Na
c. Ne
d. O

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58. $S_8$ disproportionate in alkaline medium as follows:

$$S_8(s) + 12OH^-(aq) \longrightarrow 4S^{2-}(aq) + 2X + 6H_2O(l)$$

X in the above reaction is

a. $SO_2$
b. $SO_4^{2-}$
c. S
d. $S_2O_3^{2-}$

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59. Aluminium reacts with aqueous alkali (NaOH) to

60. Which of the given elements of group 13 has the

59. Aluminium reacts with aqueous alkali (NaOH) to form

a. Aluminium oxide and water
b. Aluminium oxide and hydrogen
c. Sodium tetrahydroxoaluminate(III) and hydrogen
d. Sodium tetrahydroxoaluminate(III) and water

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60. Which of the given elements of group 13 has the highest atomic radius?

a. B
b. Al
c. Ga
d. In

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61. Consider the following statements marked as Assertion (A) and Reason (R).

Assertion (A) : Melting point of aluminium is higher than that of Boron.
Reason (R) : Aluminium has more metallic nature than that of Boron.

In the light of above statements, Choose the correct statement.

a. Both (A) & (R) are true and (R) is the correct explanation of (A).
b. Both (A) & (R) are true but (R) is the not the correct explanation of (A).
c. (A) is true but (R) is false
d. (A) is false but (R) is true

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62. The correct order of covalent radii of group 14 elements is

a. C < Si < Ge < Sn
b. C < Si < Sn < Ge
c. C < Ge < Si < Sn
d. C < Sn < Ge < Si

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63. Choose the incorrect statement.

a. Standard electrode potential for $Al^{3+}/Al$ is negative while it is positive for $Tl^{3+}/Tl$
b. Aluminium has high tendency to make $Al^{3+}$ ions as compared to thallium.
c. $Ti^{3+}$ is a powerful oxidising agent
d. Aluminium is less electropositive than thallium

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64. When borax is heated in a Bunsen burner flame with cobalt (II) oxide on a loop of platinum wire, the colour of bead formed is

a. Green
b. Red
c. Blue
d. Yellow

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65. The incorrect statement about carbon monoxide is

a. Pure CO can be prepared by dehydration of formic acid with conc. $H_4SO_4$ at 373 K
b. Complex of CO with haemoglobin is 300 times less stable than the oxygen haemoglobin complex
c. CO is used in the extraction of metals from their oxide ores
d. CO is a colourless water insoluble gas

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66. Oxidation number of nitrogen in $HNO_3$, $N_3H$, $NH_3$ are respectively

a. +5, +3, –3
b. +5, –1/3, –3
c. +3, –1/3, –3
d. +5, –1/3, +3

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67. Electrode potential for the following half cell reactions are

$Mg \longrightarrow Mg^{2+} + 2e^-$, $E° = +2.36$ V

$Ag \longrightarrow Ag^{+} + e^-$, $E° = -0.80$ V

68. Select the reaction in which $H_2O_2$ is acting as reducing agent?

a. $Mn^{2+} + H_2O_2 \longrightarrow Mn^{4+} + 2OH^-$
b. $I_2 + H_2O_2 + 2OH^- \longrightarrow 2I^- + 2H_2O + O_2$
c. $PbS + 4H_2O_2 \longrightarrow PbSO_4 + 4H_2O$

The EMF for the cell reaction

$Mg(s) + Ag^+(aq) \rightarrow Mg^{2+}(aq) + Ag(s)$ will be

a. 1.56 V
b. –1.56 V
c. 3.16 V
d. –3.16 V

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d. $2Fe^{2+} + 2H^+ + H_2O_2 \longrightarrow 2Fe^{3+} + 2H_2O$

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69. The oxide, which cannot act as a reducing agent is

a. $N_2O_5$
b. $ClO_2$
c. $SO_2$
d. $NO_2$

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70. Oxidation number of middle carbon of $C_3O_2$ is

a. 0
b. +4
c. –2
d. +2

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71. Number of moles of $Cr_2O_7^{2-}$ required to oxidize one mole of ferrous oxalate completely in acidic medium will be

a. 2
b. $\frac{1}{3}$
c. $\frac{1}{2}$
d. $\frac{1}{6}$

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72. Select the reaction which is not redox among the following.

a. $TiCl_4(l) + 2Mg(s) \longrightarrow Ti(s) + 2MgCl_2(s)$
b. $Cr_2O_3(s) + 2Al(s) \longrightarrow Al_2O_3(s) + 2Cr(s)$
c. $2H_2O(l) + F_2(g) \longrightarrow 4HF(aq) + O_2(g)$
d. $AgNO_3(aq) + KI(aq) \longrightarrow AgI(s) + KNO_3(aq)$

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73. Combination redox reaction among the following is ??

a. $2KClO_3 \rightarrow 2KCl + 3O_2$
b. $3Mg + N_2 \rightarrow Mg_3N_2$
c. $Cr_2O_3 + 2Al \rightarrow Al_2O_3 + 2Cr$
d. $Mg + 2H_2O \rightarrow Mg(OH)_2 + H_2$

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74. In the following reaction,

$2HgCl_2 + SnCl_2 \rightarrow Hg_2Cl_2 + SnCl_4$

The oxidation number of mercury changes from

a. + 2 to + 4
b. + 4 to + 2
c. + 2 to + 1
d. + 2 to zero

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75. The disproportionation reaction among the following is

a. $2F_2 + 2OH^- \longrightarrow 2F^- + OF_2 + H_2O$
b. $CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$
c. $Cl_2 + OH^- \longrightarrow ClO^- + Cl^- + H_2O$
d. $Fe + 3H_2O \longrightarrow Fe_2O_3 + 3H_2$

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76. In iodometric titration reaction, the indicator used to detect end point of the reaction is,

a. Methyl orange
b. Starch
c. Phenolphthalein
d. Methyl red

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77. Among 14 group elements, catenation tendency is minimum for

a. Si
b. Ge
c. Pb
d. Sn

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78. **Statement-I:** Boron shows diagonal relationship with silicon.
**Statement-II**: Maximum covalency of boron is three.
In the light of above statements choose the correct option among the following.

a. Both statement-I and statement-II are correct
b. Both statement-I and statement-II are incorrect
c. Statement-I is correct but statement-II is incorrect
d. Statement-I is incorrect but statement-II is correct

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79. $B_2H_6 + NH_3 \rightarrow A \xrightarrow{\Delta} B$ .

The product B is

a. $(BN)_x$
b. $2BH_3.NH_3$
c. $[BH_2(NH_3)_2]^+ [BH_4]^-$
d. $B_3N_3H_6$

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80. Number of B – O – B bonds in borax is

a. Two
b. Three
c. Four
d. Five

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81. All the tetrachlorides of group 14 elements gets easily hydrolysed by water, except

a. $SiCl_4$
b. $SnCl_4$
c. $GeCl_4$
d. $CCl_4$

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82. Match the given lists I and II and choose the correct option

| Lists I | Lists II |
|---|---|
| (a) $PbO_2$ | (i) Neutral |
| (b) GeO | (ii) Amphoteric |
| (c) CO | (iii) Basic |
| (d) $Tl_2O_3$ | (iv) Acidic |

a. (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
b. (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)
c. (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
d. (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

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83. For which of the following elements, +1 oxidation state is more stable than +3?

a. B
b. Al
c. In
d. Tl

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84. Thermodynamically most stable allotrope of carbon is/are

a. Diamond
b. Graphite
c. Fullerene
d. All are equally stable

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85. The mixture, which is called as producer gas is

a. $CO + H_2$
b. $N_2 + H_2$
c. $CO + N_2$

86. An example of man made silicate is

a. Zeolite
b. Asbestos
c. Cement

c. $CO + N_2$
d. $H_2O + CO$

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d. Mica

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87. In which of the following oxidation number of H is –1?

a. HCl
b. $H_2O$
c. $CH_4$
d. $CaH_2$

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88. Consider the following statements:

Statements I : The crystal of $SiO_2$ is considered as a giant molecule in which eight membered rings are formed with alternate Si and O atoms.

Statement II : In three dimentional network structure of $SiO_2$ each silicon atom is covalently bonded with four oxygen atoms in a tetrahedral manner

Choose the correct option.

a. Both the statements I and II are correct
b. Both the statements I and II are incorrect
c. Statements I is correct but statement II is incorrect
d. Statements II is correct but statement I is incorrect

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89. Element of group 14 which decomposes steam to form its dioxide and dihydrogen gas is

a. Carbon
b. Silicon
c. Germanium
d. Tin

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90. An aqueous solution of silver is when electrolysed using platinum electrodes then the product obtained at anode is

a. $O_2$ gas
b. $NO_2$ gas
c. $N_2$ gas
d. $N_2O$ gas

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91. All of the given metals show disproportionation except:

a. Potassium
b. Iron
c. Chromium
d. Maganese

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92. Which of the following species can function both as oxidizing as well as reducing agent?

a. $Cl^-$
b. $ClO_4^-$
c. $ClO^-$
d. $MnO_4^-$

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93. Which of the following is a decomposition redox reaction?

a. $3\,Mg + N_2 \longrightarrow Mg_3N_2$
b. $Fe + S \longrightarrow FeS$
c. $CaCO_3 \rightarrow CaO + CO_2$
d. $2\,H_2O \longrightarrow 2\,H_2 + O_2$

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94. In a balanced equation
$H_2SO_4 + xHI \rightarrow H_2S + yI_2 + zH_2O$

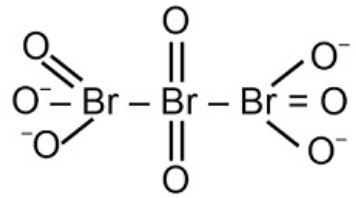
The values of x, y, z are

a. x = 3, y = 5, z = 2
b. x = 4, y = 8, z = 5
c. x = 8, y = 4, z = 4
d. x = 5, y = 3, z = 4

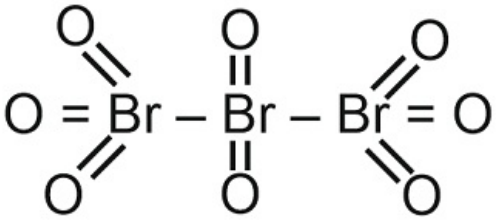
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95. The correct structure of tribromooctaoxide is

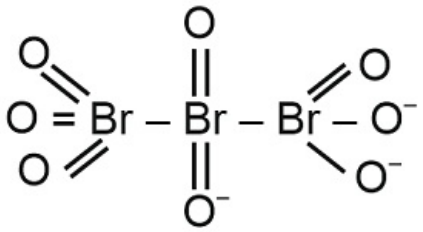
a.



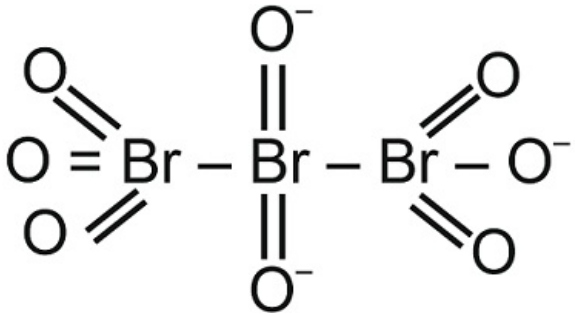
b.



c.



d.



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96. Fractional oxidation number is found in

a. $Fe_3O_4$
b. $PbO_2$
c. $Fe_2O_3$
d. PbO

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97. Total number of five membered rings in $C_{60}$ is

a. Zero
b. 2
c. 6
d. 12

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98. Covalency of Al in $[AlCl(H_2O)_5]^{+2}$ is

a. 2
b. 5
c. 4
d. 6

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99. If oxidation states of A, B and C respectively are +2, +3 and –2 then the possible formula of the compound is

a. $A_3(B_2C)_2$
b. $A_2(BC)_3$
c. $B_3(AC_3)_2$
d. $A_3(BC_3)_2$

100. Oxidation number of chlorine in $CaOCl_2$ are

a. +1, +1
b. +1, – 1
c. +1, +3
d. +1, –3

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101. Integral form of the exponential growth equation is

a. $\frac{dN}{dt} = rN$
b. $N_t = N_0 e^{rt}$
c. $\frac{dN}{dt} = (b - d) \times N$
d. $\frac{dN}{dt} = rN(1 - \frac{N}{K})$

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102. Partitioning of resources among interacting species promotes

a. Competitive co-existence
b. Competitive exclusion
c. Commensalism
d. Amensalism

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103. In a newly colonised habitat, which factor contributes more significantly for population growth?

a. Natality
b. Mortality
c. Immigration
d. Emigration

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104. Pericycle forms semi lunar patches of sclerenchyma cells in

a. Dicot roots
b. Monocot stems
c. Dicot stems
d. Monocot roots

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105. Heartwood and sapwood both are

a. Types of secondary xylem
b. Involved in conduction of water from root to leaf
c. Involved in storage of tannins and resins
d. Composed of lignified living cells

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106. Hardness of seed coats of legumes is due to

a. Sclereids
b. Collenchyma
c. Parenchyma
d. Sclerenchyma fibres

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107. *Wolffia* is included in angiosperms because

a. Seeds are not protected by seed coat
b. Seeds are enclosed in fruits
c. Of the production of two type of spores
d. Of the presence of autotrophic mode of nutrition

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108. Pteridophytes

a. Lack vascular tissues
b. Have plant body differentiated into root, stem and leaves
c. Have dominant gametophytic phase in their life cycle
d. Do not require water for their fertilisation

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109. Find the **mismatched** pair w.r.t. plants.

a. Bryophyta – Embryophytes
b. Pteridophyta – First tracheophytes
c. Gymnosperms – Naked seeded plants
d. Angiosperms – Archegoniate spermatophytes

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110. Select the **odd** one out w.r.t. heterosporous species

a. *Selaginella*
b. *Salvinia*
c. *Cedrus*
d. *Funaria*

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111. Read the following statements and select the **correct** option.

**Statement A:** The productivity and distribution of plants is also heavily dependent on water.

**Statement B:** Aquatic organism do not face any water-related problems.

a. Only statement A is correct
b. Only statement B is correct
c. Both statements are correct
d. Both statements are incorrect

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112. How many of the given factors affect(s) activities of animals?

(a) Light intensity

(b) Water pH

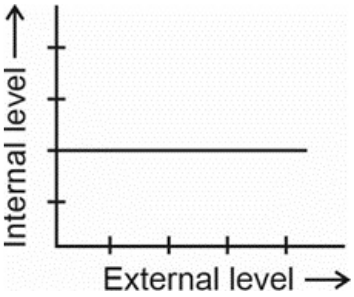
(c) Light duration

(d) Sediment characteristic

a. 3
b. 4
c. 2
d. 1

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113. Below given organismic response is **not** true for




a. Humans
b. Mammals
c. All birds
d. Plants

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114. State the true **(T)** or false **(F)** for the following statements and select the **correct** option.

(A) Success of mammals is largely due to their ability to maintain a constant body temperature.

(B) Heat loss or heat gain is a function of volume not the surface area.

(C) Invertebrates are conformers.

| | **A** | **B** | **C** |
|---|---|---|---|
| (1) | T | T | T |
| (2) | T | F | F |
| (3) | F | F | T |
| (4) | T | F | T |

a. (1)
b. (2)
c. (3)
d. (4)

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115. Thick-walled spores are formed to survive in unfavourable conditions in all, **except**.

a. Bacteria
b. Fungi
c. Lower plants
d. snails

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116. Which of the following statements is **not** true regarding axillary buds?

a. They are present in the axil of leaves
b. They are capable of forming a branch or a flower
c. They are involved in secondary growth
d. They are meristematic cells

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117. All of the following are secondary meristems **except**

a. Interfascicular cambium
b. Cork cambium
c. Intrafascicular cambium
d. Intercalary meristem

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118. Which of the given features is not true for a tracheid?

a. Tube like
b. Composed of many fused cells
c. Lignified wall
d. Tapering ends

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119. Select the **odd** one w.r.t. feature of the tissue involved in radial conduction of water

a. Suberin deposition
b. Living
c. Cellulosic
d. Thin-walled

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120. Read the following statements and select the **correct** option.

**Statement A:** Phloem fibres are made up of sclerenchymatous cells.

**Statement B:** Bast fibres are generally absent in secondary phloem.

a. Only statement A is correct
b. Only statement B is correct
c. Both statements are correct
d. Both statements are incorrect

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121. The ploidy level of protonema, spore mother cell and rhizoidal cells respectively in *Funaria* are ?

a. N, 2N, N
b. N, N, N
c. 2N, N, 2N
d. N, 2N, 2N

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122. Independent, free-living, photosynthetic gametophyte is **not** found in

a. Algae and Bryophytes
b. Gymnosperms and Angiosperms
c. Pteridophytes and Bryophytes
d. Algae and Angiosperms

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123. Find the **odd** one out w.r.t. life cycle patterns

a. In haplontic life cycle, gametophyte is dominant stage
b. In diplontic life cycle, sporophyte is dominant stage
c. In haplo-diplontic life cycle, both sporophyte and gametophyte stages are absent
d. In diplontic life cycle, sporophyte stage is multicellular

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124. If in a pond there are 80 lotus plants last year and through reproduction 8 new plants are added, taking the current population to 88. The value of birth rate per lotus per year will be

a. 11
b. 0.1
c. 0.9
d. 1.1

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125. Organism that regulates body temperature by behavioural means is

a. Antarctic fish
b. Desert lizard
c. Cactus
d. Archaebacteria

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126. Which of the following statements is **correct** w.r.t. temperature?

a. Average temperature on land is constant in every season always
b. It decreases progressively from the poles towards the equator
c. It increases from plains to the mountain tops
d. Level of thermal tolerance of different species determine their geographical distribution

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127. State **true (T)** or **false (F)** to the following statements and select the **correct** option.

A. When bulliform cells are flaccid, they make the leaves curl inwards to minimize water loss.

B. In dicot leaves, the size of the vascular bundles is dependent on the size of the veins.

C. The adaxial epidermis bears more stomata than the abaxial epidermis in dicot leaves.

| | A | B | C |
|---|---|---|---|
| (1) | T | F | T |
| (2) | F | F | T |
| (3) | T | T | F |
| (4) | F | T | T |

a. (1)
b. (2)
c. (3)
d. (4)

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128. Tissue is a group of cells which
(a) Perform a common function.
(b) Are structurally always similar.
(c) Have a common origin.

a. Only (a) and (b) are correct
b. Only (a) and (c) are correct
c. All (a), (b) and (c) are correct
d. All are correct except (a)

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129. During secondary growth in stem, after continued activity of vascular cambium

a. Primary xylem remains more or less intact
b. Secondary xylem and phloem gets crushed
c. More secondary phloem is formed as compare to secondary xylem
d. Primary phloem remains intact

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130. Allen's rule states that

a. Mammals from colder climates have longer ears and limbs to maximise heat loss
b. Mammals from colder climates have shorter ears and limbs to minimise heat loss.
c. Mammals from tropical climates have shorter ears and limbs to minimise heat loss
d. Warm blooded animals have larger body size in cold climate than in hotter area

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131. Which classification system assumes that organisms belonging to same taxa have a common ancestor?

a. Artificial system
b. Phylogenetic system
c. Natural system
d. Sexual system

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132. *Polysiphonia* reproduce asexually by

a. Non-motile spores
b. Non-motile gametes
c. Fragmentation
d. Motile spores

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133. Read the following statements and select the **correct** option.

**Statement A:** Green algae have a rigid cell wall made of an inner layer of pectose and an outer layer of cellulose.

**Statement B:** Pyrenoids are storage bodies that store food in form of oil droplets only.

134. Brown algae are not

a. Marine water forms
b. Heterotrophic
c. Filamentous
d. Coloured

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a. Only statement A is correct
b. Only statement B is correct
c. Both statements are correct
d. Both statements are incorrect

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135. Choose the odd one for economic importance of bryophytes

a. Food for herbaceous mammals
b. Can be used as packing material
c. Mainly used as ornamental plants
d. Food for birds

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136. Among the following biomes, mean annual rainfall will be maximum in

a. Temperate forest
b. Coniferous forest
c. Grassland
d. Tropical rain forest

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137. Thalloid main plant body is characteristic feature of all, **except**

a. *Marchantia*
b. *Selaginella*
c. *Funaria*
d. *Polytrichum*

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138. Prothallus represents the

a. Creeping, filamentous gametophyte of mosses
b. Inconspicuous, mostly photosynthetic and thalloid gametophytes in pteridophytes
c. Haploid spores of ferns
d. Archegonia of bryophytes

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139. Read the following **assertion** (A) and **reason** (R) and select the **correct** option

A : Trichomes are branched unicellular appendages arising from root.

R : Trichomes facilitates water loss by transpiration.

a. Both Assertion & Reason are true and the reason is the correct explanation of the assertion
b. Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
c. Assertion is true statement but Reason is false
d. Both Assertion and Reason are false statements

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140. Pericycle is **not** seen in

a. Monocot stems and dicot stems
b. Monocot roots and dicot roots
c. Monocot stems only
d. Dicot stems only

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141. Select the **incorrect** statement regarding bryophytes.

a. Sex organs are multicellular
b. Antheridium produce biflagellate antherozoids
c. Archegonium is flask shaped and produce many eggs.
d. Gametophyte produce gametes

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142. Haploid spores in bryophytes germinate to produce

a. Free-living gametophyte
b. Diploid Gametophyte
c. Free-living sporophyte
d. Sporophyte attached to the parent plant

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143. Antheridia and archegonia in mosses are

144. Highly reduced male gametophyte in

produced at

a. Prothallus
b. Protonema
c. Leafy shoot
d. Sporophylls

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gymnosperms called

a. Anther
b. Antheridium
c. Antherozoid
d. Pollen grain

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145. Parenchymatous cells lie between xylem and phloem patches in dicot roots are called

a. Pericycle
b. Conjunctive tissue
c. Endodermis
d. Pith

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146. Phloem parenchyma is generally absent in

a. Monocot stem
b. Dicot stem
c. Dicot leaf
d. Dicot root

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147. Select the **incorrect** statement.

a. Vascular cambium is completely secondary in origin in dicot root
b. Bark is a technical term that refers to all tissue interior to vascular cambium
c. Lenticels permit the exchange of gases
d. Bark that is formed early in the season is called soft bark

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148. Choose the **correct** option regarding given statements.

**Statement A:** Polar aquatic mammals like seals have a thin layer of fat (blubber) below their skin.

**Statement B:** Polar aquatic mammals are not able to maintain homeostasis.

a. Only statement A is correct
b. Only statement B is correct
c. Both statements are correct
d. Both statements are incorrect

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149. Select the **odd** one w.r.t. predation.

a. Tiger and Deer
b. Sparrow and seeds
c. Herbivores and plants
d. Lichens

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150. Which of the following is the **odd** w.r.t. parasites.

a. Female mosquito
b. *Plasmodium*
c. Lice
d. *Cuscuta*

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151. Specialised cells in our body like macrophages and leucocytes exhibit the 'X' type of movement. This 'X' type of movement is also seen in

a. *Euglena*
b. *Hydra*
c. *Amoeba*
d. *Paramoecium*

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152. Unmyelinated nerve fibre possesses all of the following, **except**

a. Axoplasm
b. Schwann cells
c. Neurilemma
d. Myelin sheath

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153. The common feature between the type of muscle

154. Each organised skeletal muscle in our body is made of a number of muscle bundles held together by

fibres present in the wall of stomach and the type of muscle fibres present in the heart is

a. Presence of striations
b. Absence of branching pattern
c. Presence of cylindrical shaped muscle fibres
d. Involuntary action

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middle of a number of muscle bundles held together by a common _____ connective tissue layer. Select the correct option to fill in the blank.

a. Chitinous
b. Collagenous
c. Calcareous
d. Phosphorylated

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155. Choose the incorrect statement w.r.t. a sarcomere.

a. 'I' band consists of only light bands made up of actin filaments
b. 'A' band consists of only myosin filaments
c. 'H' zone is the central part of thick filament not overlapped by thin filament.
d. 'Z' line is an elastic fibre that bisects 'I' band

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156. The active ATPase enzyme and binding sites for ATP are present on

a. Short arm of meromyosin
b. Light meromyosin
c. Cross arm of light meromyosin
d. Globular head of meromyosin

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157. Mechanism of muscle contraction is best explained by the sliding filament theory. Select the incorrect statement w.r.t. it.

a. Breaking of crossbridge occurs when the myosin head binds to ATP.
b. Myosin head bound with ADP and Pi is considered as energised state.
c. Formation of crossbridge occurs when $Ca^{2+}$ level in the sarcoplasm is low and myosin head is not attached with ATP
d. During muscle contraction, actin and myosin filaments do not shorten but actin filaments slide over myosin filaments

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158. When the muscle is in maximally contracted state, then which of the given events occur?

(a) Reduction in length of 'I' band

(b) Widening of 'A' band

(c) Disappearance of 'H' zone

(d) Utilisation of the energy released from hydrolysis of ATP into ADP + $P_i$ by myosin

Choose the **correct** option

a. a, b, c and d
b. a, b and c only
c. b, c and d only
d. a, c and d only

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159. The total number of facial bones in an adult human is

a. Equal to the total number of true ribs
b. Less than the number of thoracic vertebrae
c. Thrice the number of vertebrochondral ribs
d. Equal to the number of carpals in both forelimbs

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160. Select the correct matching of the type of the joint with its example in human skeletal system .

a. Fibrous joint-Between $1^{st}$ and $2^{nd}$ cervical vertebrae
b. Cartilaginous joint-Between temporal and occipital bone
c. Saddle joint-Between carpal and metacarpal of thumb
d. Hinge joint-Between femur and pelvis

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161. Choose the correct statement regarding the disorders of muscular or skeletal system.

a. Rapid spasms in muscle due to high $Ca^{2+}$ in body

162. When a neuron is not conducting any impulse, the axonal membrane is

a. Completely Impermeable to $Na^{+}$ ions

fluid causes tetany
b. Osteoporosis is characterised by decrease in bone mass and higher chances of fractures with advancing age.
c. Myasthenia gravis is a genetic disorder in which there is inhibition of sliding of myosin filaments
d. Muscular dystrophy is an auto-immune disorder characterised by shortening of muscles.

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b. Permeable to negatively charged proteins present in the axoplasm
c. Positive on inner surface while negative on outer surface
d. More permeable to $K^+$ ions

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163. A person 'X' was analysing the different types of neurons. He found that the type of neurons present majorly in the cerebral cortex of humans possess cell body with

a. One axon and no dendrites
b. One axon and two or more dendrites
c. One axon and one dendrite
d. No dendrites and two axons

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164. The somatic neural system relays impulses from CNS to muscles attached to/present in __A__ while the autonomic neural system transmits impulses from the CNS to the muscles found in __B__

Select the option that correctly identifies 'A' and 'B' respectively

a. Bone, Heart
b. Heart, Iris of eyeball
c. Stomach, Oesophagus
d. Heart, Small intestine

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165. If a comparison is made between nerve cells and muscle cells, then the latter can be differentiated from the former by the presence of all of the given properties, **except**

a. Contractility
b. Extensibility
c. Excitability
d. Elasticity

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166. Three major regions *i.e.* 'X', 'Y' and 'Z' make up the brain stem. 'Y' and 'Z' are the parts of hindbrain. Identify 'X' and choose the incorrect statement w.r.t. it.

a. 'X' is located between the thalamus/hypothalamus of the forebrain and pons of the hindbrain
b. A canal called cerebral aqueduct passes through the 'X'
c. The dorsal portion of 'X' consists of corpora quadrigemina
d. 'X' contains centres which control respiration, cardiovascular reflexes and body temperature

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167. How many of the hormones given in the box below are considered as trophic hormones?

Prolactin, Oxytocin, Vasopressin, Luteinizing hormone, Follicle stimulating hormone

a. Four
b. Three
c. Two
d. One

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168. The hormone whose hypersecretion is associated with Graves' disease is involved in

a. Regulating the blood calcium levels
b. Maintaining the normal rhythms of sleep-wake cycle
c. Metabolism of carbohydrates, proteins and fats
d. Differentiation of T-lymphocytes

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169. **Assertion (A) :** Red muscle fibres are also known as aerobic muscles.

**Reason (R) :** Red muscle fibres utilise the large amount of oxygen stored in them for ATP production.

170. Select the incorrect statement w.r.t. an adult human.

a. The skull region articulates with the superior region of the vertebral column with the help of

In the light of above statements, select the correct option.

a. Both (A) and (R) are true, (R) is the correct explanation of (A)
b. Both (A) and (R) are true, but (R) is not the correct explanation of (A)
c. (A) is true, (R) is false
d. Both (A) and (R) are false

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two occipital condyles
b. Sternum is the flat bone on the ventral midline of the thorax
c. Appendicular skeleton comprises 126 bones and consists of bones of the limbs along with their girdles
d. Scapula is situated in the ventral part of the thorax between the second and the seventh ribs

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171. Select the **incorrect** statement regarding mechanism of hormone action.

a. Estrogen is a lipid-soluble hormone capable of diffusing inside the cell.
b. Glucagon is a peptide hormone which does not enter the target cell.
c. Epinephrine attaches to the receptor present on the target cell surface.
d. Iodothyronines interact with membrane bound receptors to regulate gene expression.

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172. Glucocorticoids promote all, **except**

a. Gluconeogenesis
b. Lipolysis
c. Proteolysis
d. Storage of glucose in liver

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173. Select the option which is **not** a direct function of gonadotropins.

a. Androgen secretion from testis
b. Maintenance of corpus luteum
c. Ovulation
d. Maintenance of secondary sexual characters only in males

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174. While pituitary dwarfs and cretins are somewhat of same height, the main difference is that

a. Head of cretin is larger
b. Cretin dwarfs have long extremities
c. Pituitary dwarfs have normal intelligence while cretins do not
d. Pituitary dwarfs are with dark complexion

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175. A person passes excessive urine and drinks much water but his glucose level in blood is normal. It is due to

a. Increased secretion of glucagon
b. Fall in somatostatin level in blood
c. Increase in insulin secretion
d. Reduction in vasopressin secretion

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176. **Assertion (A) :** Erythropoietin hormone increases stem cell mitosis and speeds up development of RBCs.

**Reason (R) :** Kidneys do not play any role in the formation of RBCs.

In the light of above statements, select the correct option.

a. Both Assertion and Reason are true and the reason is the correct explanation of the Assertion
b. Both Assertion and Reason are true but the reason is not the correct explanation of the Assertion
c. Assertion is true statement but Reason is false
d. Both Assertion and Reason are false statements

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177. The immune responses of old persons become weak due to degeneration of

178. Select a peptide hormone, which acts mainly on hepatocytes and stimulates glycogenolysis resulting in hyperglycemia

a. Thymus gland
b. Pineal gland
c. Thyroid gland
d. Adrenal gland

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a. Insulin
b. Glucagon
c. Cortisol
d. Adrenaline

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179. Which of the following statements is **incorrect** w.r.t. hypothalamus?

a. Hypothalamus is the basal part of diencephalon of forebrain
b. It contains several groups of neurosecretory cells called nuclei which produce hormones
c. Hypothalamic hormone GnRH acts on gonads and stimulates the synthesis and release of gonadotrophins
d. The posterior pituitary is under the direct neural regulation of the hypothalamus

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180. All of the following are functions of progesterone, **except**

a. It supports pregnancy
b. Stimulates the formation of alveoli
c. Regulates female sexual behaviour
d. Stimulates milk secretion

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181. The greyish appearance of cerebral cortex is largely due to

a. Accumulation of myelin sheath
b. Concentrated neuron cell bodies
c. Presence of iron containing cytochrome pigment
d. Presence of myelinated axon

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182. The $Na^+$ – $K^+$ pump transports 3 __A__outwards for 2 __B__into the axon maintaining the polarisation state when the neuron is in __C__.
Choose the option that **correctly** fills the blanks A, B and C.??

a. $K^+$, $Na^+$, Depolarisation state
b. $K^+$, $Na^+$, Repolarisation state
c. $Ca^{+2}$, $K^+$, Repolarisation state
d. $Na^+$, $K^+$, Resting state

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183. Select the option with correct set of bones which exhibits same number of bones

a. Metacarpals of forelimbs; metatarsals of hind limbs
b. Cranial bones; facial bones
c. Pectoral girdle; pelvic girdle
d. Carpals of forelimbs, tarsals of hind limbs

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184. On the basis of following characters, identify 'A' and 'B' in a human.
A. Pliable matrix due to chondroitin salts
B. Hard matrix due to calcium salts

Select the correct option.

a. A-Cartilage present at the joint between $8^{th}$ and $7^{th}$ rib, B-Sternum
b. A-Fibula, B-Sternum
c. A-Hyaline cartilage, B-Fibrous cartilage
d. A-Fibula, B-Pubic symphysis

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185. The type of muscle fibres present in our

a. Biceps are uninucleated and voluntary
b. Heart are non-striated and involuntary
c. Thigh are multinucleated and fusiform in shaped
d. Stomach are unbranched and involuntary

186. Select the **correct** match w.r.t. human bone and its feature

a. Hyoid – 'L' shaped bone
b. Ribs – Bicephalic
c. Coxal bone – Acromion process
d. Sternum – Neural canal

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187. **Assertion (A) :** During vigorous exercise, muscles lead to fatigue and pain.

**Reason (R) :** The muscle does not get enough oxygen and undergoes anaerobic respiration to produce lactic acid.

In the light of above statements, select the correct option.

a. Both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion
b. Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion
c. Both Assertion and Reason are false statements
d. Assertion is true statement but Reason is false

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188. Synovial joints have a fluid filled cavity between the articulating surfaces of the

a. Two bones
b. Bone and tendon
c. Tendon and cartilage
d. Two muscles

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189. How many of these bones belong to lower limb in humans?

| Parietal, Occipital, Temporal, Tibia, Femur, Patella, Fibula, Radius, Ulna |
|---|

a. Six
b. Four
c. Seven
d. Nine

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190. Layer of retinal cells that responds first to 'light' as stimulus is composed of

a. Bipolar cells
b. Multipolar cells
c. Ganglionic cells
d. Rods and cones

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191. How many structures from given below in the box are present in the neural system of both humans and insects?

| Brain, Ganglia, Neural tissue, Fibroblasts |
|---|

a. One
b. Two
c. Three
d. Four

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192. Select the **incorrect** statement w.r.t. parathyroid gland and its secretion.

a. PTH is a hypercalcemic hormone and can maintain $Ca^{2+}$ homeostasis.
b. Parathyroid hormone is a steroid hormone that gets activated when there is low $Ca^{2+}$ level in blood.
c. PTH may lead to activation of cells responsible for dissolution of bones.
d. PTH stimulates reabsorption of $Ca^{2+}$ by the renal tubules.

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193. Primary function of which gland is related to biological rhythms?

a. Thyroid
b. Pineal gland
c. Pituitary gland
d. Adrenal cortex

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194. If a heart transplanted patient receives glucocorticoid to help prevent rejection of transplanted organ, what effect will be observed on ACTH (Adrenocortico tropic hormone) and CRH (corticotropin releasing hormone)?

| | **ACTH level** | **CRH level** |
|---|---|---|
| (1) | High | Low |
| (2) | Low | High |

(2) Low High
(3) Low Low
(4) High High

a. (1)

b. (2)

c. (3)

d. (4)

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195. Thyroxine hormone does **not** act as

a. Intelligence hormone
b. Energy producing agent by increasing BMR
c. An agent to maintain the electrolyte balance
d. Emergency hormone

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196. Set of hormones that interact with membrane bound receptor of target cells include

a. Vasopressin and estrogen
b. FSH and GnRH
c. Adrenaline and Testosterone
d. Glucagon and progesterone

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197. Thyroid gland secretes thyrocalcitonin which regulates the blood calcium levels. Chemical nature of thyrocalcitonin is

a. Proteinaceous
b. Iodothyronines
c. Steroid
d. Amino-acid derivative

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198. Under the influence of which hormone, interstitial cells in testes secrete androgens?

a. FSH
b. LH
c. GH
d. ACTH

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199. Read the given statements and choose the correct option.

**Statement A** : Reissner's membrane separates scala vestibuli and scala media while basilar membrane separates scala media and scala tympani.

**Statement B** : Scala media is filled with endolymph while scala vestibuli and scala tympani are filled with perilymph.

a. Both the statements are correct
b. Both the statements are incorrect
c. Only statement (A) is correct
d. Only statement (B) is correct

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200. In case of knee-jerk reflex, the cyton of sensory neuron is present in (A) while the cyton of motor neuron is present in (B)

Choose the option that correctly identifies 'A' and 'B' respectively.

a. Motor endplate, gray matter of spinal cord
b. Dorsal root ganglion, gray matter of spinal cord
c. Motor endplate, white matter of spinal cord
d. Muscle spindle, gray matter of spinal cord

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